Portfolio Optimization Using Efficient Frontier

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1 Objective :

To construct an optimized investment portfolio using the Modern Portfolio Theory (MPT) framework. The goal is to identify the efficient frontier and determine the portfolio that offers the maximum Sharpe ratio, indicating the best risk-adjusted return.

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2 Methodology :

1. Data Source: Historical closing prices of 5 major Indian banking stocks listed on the NSE: HDFCBANK.NS, ICICIBANK.NS, SBIN.NS, KOTAKBANK.NS, AXISBANK.NS
2. Time Period: From January 1, 2023 to January 1, 2025.
3. Tools Used: Python packages: numpy, pandas, matplotlib, yfinance
4. Steps Involved:
    - Download stock price data and compute daily returns.
    - Calculate annualized returns and covariance matrix.
    - Use Monte Carlo simulation to generate 100,000 random portfolios.
    - For each portfolio, compute: Expected return, Risk (standard deviation), Sharpe ratio.
    - Identify the portfolio with the highest Sharpe ratio.
    - Plot the efficient frontier and highlight the optimal portfolio.

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3 Key Findings :

1. The optimal portfolio had the following characteristics:
    - Return: highest among all risk-adjusted portfolios
    - Risk: relatively low standard deviation for return achieved
    - Sharpe Ratio: maximized
2. The weight distribution of the optimal portfolio:
    - SBIN.NS - 72.1%
    - AXISBANK.NS - 21.2%
    - ICICIBANK.NS - 3.9%
    - KOTAKBANK.NS - 1.5%

- HDFCBANK.NS - 1.3%

3. The efficient frontier visualization clearly showed the trade-off between risk and return and the position of the optimal portfolio as the "tangency point."

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4 Conclusion :

This project demonstrates a practical application of portfolio optimization using historical data and the efficient frontier concept. The Monte Carlo simulation effectively explores the risk-return space, while the Sharpe ratio helps in identifying the best-performing portfolio. This methodology provides a strong foundation for constructing diversified portfolios based on quantitative techniques.

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5 Code and analysis

Libraries:

[1]:
```
import numpy as np
import matplotlib.pyplot as plt
import yfinance as yf
import pandas as pd
```

Define stock ticker set (Top 5 banks in NSE)

[2]:
```
ticker_set=["HDFCBANK.NS", "ICICIBANK.NS", "SBIN.NS", "KOTAKBANK.NS", "AXISBANK.
↪NS"]
```

Define time period for analysis

[3]:
```
start_date = "2023-01-01"
end_date = "2025-01-01"
```

Download stock data

[4]:
```
stock_data = yf.download(ticker_set, start=start_date, end=end_date)['Close']
stock_return = stock_data.pct_change().dropna()
```

```
YF.download() has changed argument auto_adjust default to True

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```

Compute annualized returns and covariance matrix

[5]:
```
returns = stock_return.mean() * 252
cov_matrix = stock_return.cov() * 252
num_assets = len(ticker_set)
risk_free_rate = 0.03
```

Monte Carlo simulation for portfolio optimization

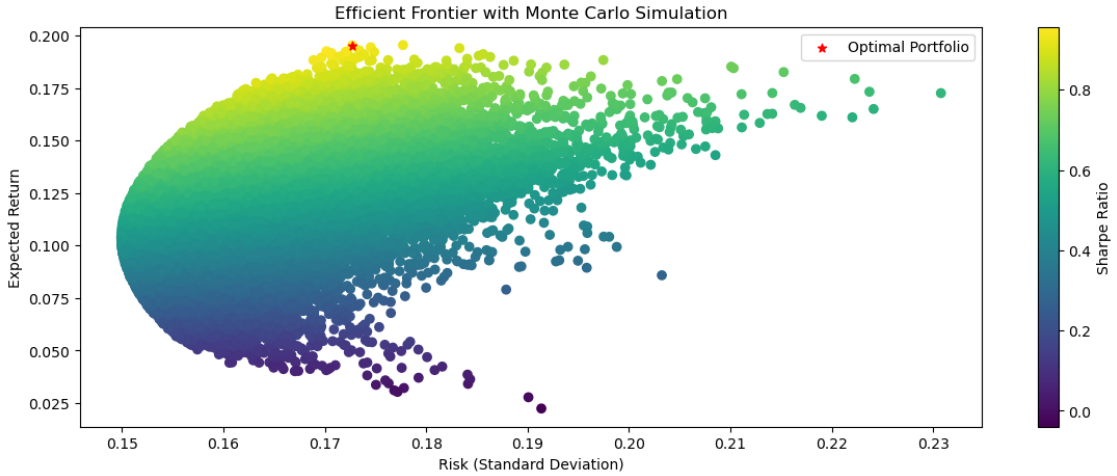
[6]:
```
num_portfolios = 10**5
all_weights = np.zeros((num_portfolios, num_assets))
ret_arr = np.zeros(num_portfolios)
vol_arr = np.zeros(num_portfolios)
sharpe_arr = np.zeros(num_portfolios)

for i in range(num_portfolios):
    weights = np.random.random(num_assets)
    weights = weights / np.sum(weights)  # Normalize so sum of weights = 1


    portfolio_return = np.dot(weights, returns)
    portfolio_variance = np.dot(weights.T, np.dot(cov_matrix, weights))
    portfolio_std = np.sqrt(portfolio_variance)


    all_weights[i, :] = weights
    ret_arr[i] = portfolio_return
    vol_arr[i] = portfolio_std
    sharpe_arr[i] = (portfolio_return - risk_free_rate) / portfolio_std
```

Find the max Sharpe Ratio portfolio

[7]:
```
max_sharpe_idx = np.argmax(sharpe_arr)
optimal_weights = all_weights[max_sharpe_idx, :]
optimal_return = ret_arr[max_sharpe_idx]
optimal_risk = vol_arr[max_sharpe_idx]
```

Plot Efficient Frontier

[8]:
```
plt.figure(figsize=(14, 5))
plt.scatter(vol_arr, ret_arr, c=sharpe_arr)
plt.colorbar(label='Sharpe Ratio')
plt.scatter(optimal_risk, optimal_return, c='red', marker='*', label='Optimal␣
↪Portfolio')
plt.xlabel('Risk (Standard Deviation)')
plt.ylabel('Expected Return')
plt.title('Efficient Frontier with Monte Carlo Simulation')
plt.legend()
plt.show()
```



Optimal Portfolio Allocation:

[9]:
```
portfolio_df = pd.DataFrame({'Stock': ticker_set, 'Weight': optimal_weights})
portfolio_df = portfolio_df.sort_values(by='Weight', ascending=False)

print("Optimal Portfolio Allocation:")
print(portfolio_df.to_string(index=False))
```

```
Optimal Portfolio Allocation:
       Stock   Weight
     SBIN.NS 0.720737
  AXISBANK.NS 0.212204
ICICIBANK.NS 0.039479
KOTAKBANK.NS 0.015034
 HDFCBANK.NS 0.012546
```

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